

Krishnaa Vadivel

GPU Scientific Software | HPC

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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Summary

GPU and scientific-software engineer with research and industry experience in distributed numerical computing, CUDA/ROCm portability, scientific ML, and performance debugging on NVIDIA and AMD systems.

Technical Skills

GPU/HPC: CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), CUDA-aware MPI, MPI, OpenMP, PETSc, SLEPc, CMake, strong/weak scaling, TotalView, Linaro DDT

Platforms: Frontier (AMD MI250X), Perlmutter (NVIDIA A100), Frontera, Linux, Docker, Kubernetes

Languages/ML: Python, C++, C, JAX, PyTorch, PyTorch Geometric, equivariant GNNs, automatic differentiation

Experience

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Engineer distributed first-principles simulation workflows in Python/C++ using MPI, OpenMP, PETSc, and SLEPc for national-lab GPU clusters.
- Port and debug multi-GPU matrix/vector workloads across CUDA and ROCm, using collective communication, scaling runs, TotalView, and Linaro DDT.
- Build DeePH-style computational graphs and train PyTorch Geometric equivariant models for LiF and double-perovskite calculations; automatic differentiation removed separate derivative calculations and delivered roughly 100x speedup in portions of the workflow.

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Implemented CUDA/OpenCL pipelines and OpenGL tooling for high-volume acoustic-waveform processing, visualization, and field diagnostics.
- Built C#/ASP.NET/SQL Server systems for remote-tool testing and calibration data workflows.

Selected Projects

Portable first-principles stack — Containerized CUDA-ready scientific software for reproducible deployment across HPC environments.

EGNN force prediction — Used PyTorch Geometric/e3nn to predict energies and obtain forces through automatic differentiation.

DOLFINx/PETSc simulations — Implemented PDE workflows for heat, electrostatics, and incompressible flow with explicit solver configurations.

Education

PhD, Electrical Engineering

Yale University, Scientific computing and computational materials

2021–present

MPhil and MS, Electrical Engineering

Yale University

2023

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Research Record

Publication: Co-author, *Journal of the American Chemical Society*, 2025.

Presentations: APS research presentations, 2024–2026, on first-principles excited-state physics.

Library Detail

Numerical stack: PETSc, SLEPc, DOLFINx, mpi4py, petsc4py, slepc4py, NumPy, SciPy, PyVista